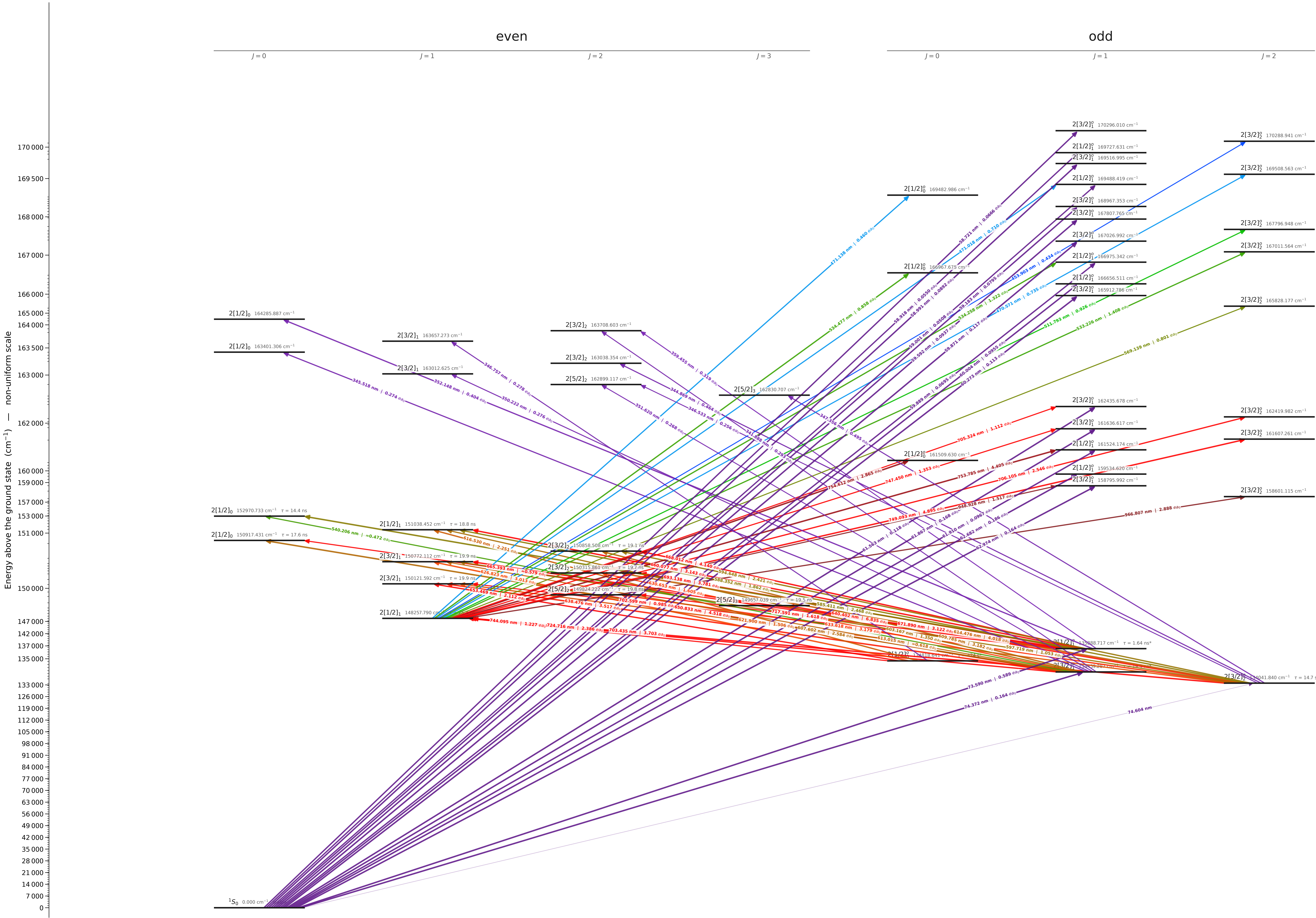
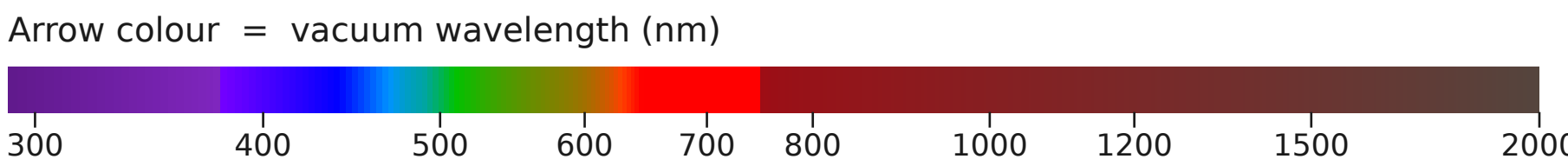


Neon

Ne (Ne I). Strongest classified transitions below 2 μm (NIST): 49 levels, 74 lines.

Ne+ ionisation limit 173929.75 cm⁻¹ = 21.56454 eV (57.494 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle U||er||J\rangle$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle U er J\rangle$ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
¹ S ₀ – 2[1/2] ₁ ^o	73.5896	–	4073.8413	0.589	5.880e+08	97.16	96.3	73.6 nm resonance line
¹ S ₀ – 2[3/2] ₁ ^o	74.3720	–	4030.9880	0.164	4.400e+07	7.978	87.8	74.4 nm resonance line
¹ S ₀ – 2[3/2] ₂ ^o	74.6036	–	4018.4733	–	6.790e-02	1.08e-08	100	74.6 nm M2 decay of the metastable 3P2 state
2[3/2] ₂ ^o – 2[3/2] ₁ ^o	614.4763	614.3063	487.8829	4.018	2.820e+07	8.079	55.5	614.3 nm line used for collinear laser spectroscopy of Ne...
2[3/2] ₂ ^o – 2[5/2] ₁	640.4017	640.2247	468.1319	6.835	5.149e+07	8.162	100	640.2 nm laser-cooling / MOT line

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

²¹Ne hyperfine structure (I = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
2[3/2] ₂ ^o	-267.68(3)	-111.55(10)	F=1/2: +1106.95 F=3/2: +803.04 F=5/2: +203.56 F=7/2: -830.93
2[5/2] ₃	-142.40(20)	-107.7(11)	F=3/2: +789.78 F=5/2: +514.55 F=7/2: +53.85 F=9/2: -667.73
2[3/2] ₁ ^o	-185.83(21)	–	F=1/2: +836.24 F=3/2: +557.49 F=5/2: +92.92 F=7/2: -557.49

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
¹ S ₀ – 2[3/2] ₁ ^o	62.974	22-20	2298(31)
¹ S ₀ – 2[1/2] ₁ ^o	73.590	22-20	-90(90)
¹ S ₀ – 2[3/2] ₂ ^o	74.372	22-20	420(90)
2[3/2] ₂ ^o – 2[1/2] ₀	540.206	22-20	1804(26)
2[1/2] ₂ ^o – 2[1/2] ₀	585.411	22-20	2286.3(27)
2[3/2] ₂ ^o – 2[1/2] ₁	588.352	22-20	1737.0(5)
2[3/2] ₂ ^o – 2[3/2] ₁	594.648	22-20	1716.0(11)
2[3/2] ₂ ^o – 2[3/2] ₁	597.719	22-20	1727.4(19)
2[3/2] ₂ ^o – 2[1/2] ₁	603.167	22-20	1744(5)
2[3/2] ₂ ^o – 2[1/2] ₀	607.602	22-20	1702.0(25)
2[3/2] ₂ ^o – 2[3/2] ₁	609.785	22-20	1725.3(25)
2[3/2] ₂ ^o – 2[3/2] ₁	613.015	22-20	1737.6(20)
2[3/2] ₂ ^o – 2[3/2] ₁	614.476	22-20	1663.0(12)
2[3/2] ₂ ^o – 2[3/2] ₁	614.476	21-20	874.94
2[1/2] ₂ ^o – 2[1/2] ₁	616.530	22-20	1656.4(28)
2[3/2] ₂ ^o – 2[3/2] ₁	621.900	22-20	1669.43(8)
2[1/2] ₂ ^o – 2[3/2] ₁	626.823	22-20	1647.38(14)
2[3/2] ₂ ^o – 2[3/2] ₁	630.653	22-20	1674.27(19)
2[3/2] ₂ ^o – 2[5/2] ₂	633.618	22-20	1641.05(16)
2[3/2] ₂ ^o – 2[3/2] ₁	638.476	22-20	1679.60(26)
2[3/2] ₂ ^o – 2[5/2] ₃	640.402	22-20	1626.05(11)
2[3/2] ₂ ^o – 2[5/2] ₃	640.402	21-20	855.7(10)

Shift = ν(first isotope) – ν(second isotope).